

# CS201 Final

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**Abstract—Write only ONE function per question in C++. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures, (iv) you can not modify the input.**

## I. QUESTION

Modify the breadth first search linked list implementation of Graph such that it checks if there is a path from the node index1 to index2.

```
bool breadthFirstSearch(bool *visit,
                        int index1,
                        int index2)
```

## II. QUESTION

Write another constructor method

```
LinkedList(LinkedList* list,
           int k,
           int m)
```

which constructs the original list from  $k$  lists of size  $m$  as follows. The first, second,  $\dots$ ,  $k$ 'th element of the original linked list will be the first element of the first, second,  $\dots$ ,  $k$ 'th input linked list, the  $k+1$ 'st,  $k+2$ 'nd,  $\dots$ ,  $2k$ 'th element of the original linked list will be the second element of the first, second,  $\dots$ ,  $k$ 'th input linked list. The elements of the original linked list should be created (not copied from the linked lists). Your algorithm should run in  $\mathcal{O}(km)$  time. Allocate and use an external array of size  $k$  to store current pointers in each of  $k$  lists.

Sample:

list[0]: a1 → a2 → a3

list[1]: b1 → b2 → b3

$k = 2, m = 3$

Output:

a1 → b1 → a2 → b2 → a3 → b3

## III. QUESTION

Write a recursive method in TreeNode class

```
void lessThanX(int* a, int& index, int x)
```

that accumulates all nodes in the binary search tree which have value less than  $x$  in array  $a$ . Use and modify index to store the integers. Your method should run in  $\mathcal{O}(\log N + K)$  time, where  $N$  is total number of nodes and  $K$  is the number of nodes which have value larger than  $x$  in the tree.

## IV. QUESTION

Modify the original selection sort so that

```
void selectionSort(LinkedList l)
```

it will use the same algorithm but sorts the elements in the linked list  $l$ . Write your method in SelectionSort class. You should swap the data not the links using setData method.

## V. QUESTION

Given an array of  $N$  numbers write a function that determines the mode of the array, i.e. the most frequent number. Your function should run in  $\mathcal{O}(N \log N)$  time. You can modify input array  $A$ . You can use sort functions in the exam library. Write your method in QuickSort class.

```
int mode(int* A, int N)
```

## VI. QUESTION

```
1 int algorithm2(int N){  
2     if (N == 1)  
3         return 0;  
4     int sum = 0;  
5     for (int i = N; i > 0; i--)  
6         for (int j = 0; j < N; j++)  
7             for (int k = 0; k < N; k++)  
8                 sum++;  
9     for (int i = 0; i < 8; i++)  
10        for (int j = 0; j < 2; j++)  
11            sum += algorithm2(N / 2);  
12    return sum;  
13 }
```

What is the time complexity of algorithm2? You must construct the recurrence equation and solve it without using the master theorem.